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class Person:
    def __init__(self, name, age):
        self.__name = name        # Encapsulation
        self.__age = age

    def get_name(self):
        return self.__name

    def get_age(self):
        return self.__age

class Patient(Person):            # Inheritance
    def __init__(self, name, age, disease):
        super().__init__(name, age)
        self.disease = disease

    def display(self):             # Polymorphism
        print(f"Patient: {self.get_name()}, Age: {self.get_age()}, Disease: {self.disease}")

class Doctor(Person):
    def __init__(self, name, age, specialization):
        super().__init__(name, age)
        self.specialization = specialization

    def display(self):
        print(f"Doctor: {self.get_name()}, Age: {self.get_age()}, Specialization: {self.specialization}")

class Appointment:
    def __init__(self, patient, doctor, date):
        self.patient = patient
        self.doctor = doctor
        self.date = date

    def display(self):
        print(f"{self.patient} {self.doctor}, Date: {self.date}")

class Hospital:
    def __init__(self):
        self.patients = []
        self.doctors = []
        self.appointments = []

    def add_patient(self):
        name = input("Patient Name: ")
        age = int(input("Age: "))
        disease = input("Disease: ")

        self.patients.append(Patient(name, age, disease))
        print("Patient Added Successfully.\n")

    def view_patients(self):
        if not self.patients:
            print("No Patients Found.\n")
        else:
            for p in self.patients:
                p.display()

    def add_doctor(self):
        name = input("Doctor Name: ")
        age = int(input("Age: "))
        specialization = input("Specialization: ")

        self.doctors.append(Doctor(name, age, specialization))
        print("Doctor Added Successfully.\n")

    def view_doctors(self):
        if not self.doctors:
            print("No Doctors Found.\n")
        else:
            for d in self.doctors:
                d.display()

    def book_appointment(self):
        patient = input("Patient Name: ")
        doctor = input("Doctor Name: ")
        date = input("Appointment Date: ")

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def book_appointment(self):
    self.appointments.append(Appointment(patient, doctor, date))
    print("Appointment Booked.\n")

def view_appointments(self):
    if not self.appointments:
        print("No Appointments.\n")
    else:
        for a in self.appointments:
            a.display()

hospital = Hospital()

while True:

    print("\n HOSPITAL MANAGEMENT SYSTEM ")
    print("1. Add Patient")
    print("2. View Patients")
    print("3. Add Doctor")
    print("4. View Doctors")
    print("5. Book Appointment")
    print("6. View Appointments")
    print("7. Exit")

    choice = input("Enter Choice: ")

    if choice == "1":
        hospital.add_patient()

    elif choice == "2":
        hospital.view_patients()

    elif choice == "3":
        hospital.add_doctor()

    elif choice == "4":
        hospital.view_doctors()

    elif choice == "5":
        hospital.book_appointment()

    elif choice == "6":
        hospital.view_appointments()

    elif choice == "7":
        print("Thank You!")
        break

    else:
        print("Invalid Choice!")

```

HOSPITAL MANAGEMENT SYSTEM

1. Add Patient
2. View Patients
3. Add Doctor
4. View Doctors
5. Book Appointment
6. View Appointments
7. Exit

Patient Added Successfully.

HOSPITAL MANAGEMENT SYSTEM

1. Add Patient
2. View Patients
3. Add Doctor
4. View Doctors
5. Book Appointment
6. View Appointments
7. Exit

Patient: Sagar, Age: 28, Disease: Fever

HOSPITAL MANAGEMENT SYSTEM

1. Add Patient
2. View Patients
3. Add Doctor
4. View Doctors
5. Book Appointment
6. View Appointments
7. Exit

Doctor Added Successfully.

HOSPITAL MANAGEMENT SYSTEM

1. Add Patient
2. View Patients
3. Add Doctor
4. View Doctors
5. Book Appointment
6. View Appointments
7. Exit

Doctor: Shivam, Age: 30, Specialization: xyz

HOSPITAL MANAGEMENT SYSTEM

1. Add Patient
2. View Patients
3. Add Doctor
4. View Doctors
5. Book Appointment
6. View Appointments
7. Exit

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